

A Nature-Inspired Distributed Robust Control Design for Ground-Aerial Vehicle Cooperation

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Abstract—This paper studies the problem of tracking a moving ground object that can be found in a number of applications such as surveillance, criminal and military situations. Ground vehicles can be employed for the tracking, but are greatly affected by surrounding obstacles, e.g. building blocks, roadside objects, other vehicles and passengers, etc. Hence, their cooperation with an aerial vehicle, e.g. a drone, which has a better vision from above, is proposed. This is inspired from the wolf-raven cooperative hunting behavior in nature, where the aerial vehicle acts as the raven and ground vehicles act as wolves. However, the position and velocity sensing as well as the vision of ground and aerial vehicles are still perturbed by the surrounding environment. Therefore, a distributed robust control design is proposed for the ground-aerial vehicle cooperation to suppress the effects of the aforementioned disturbances. Conditions on the controller parameter are derived so that the proposed controller is robust in the H_∞ or H_2 sense, depending on the disturbance types. Furthermore, the bounded input constraint, as often encountered in reality, is investigated, revealing that the proposed controller is input-to-state stable with environment disturbances. Numerical simulations are then carried out to illustrate the effectiveness of the proposed nature-inspired distributed robust controller design.

Index Terms—Ground-aerial vehicle cooperation, nature-inspired algorithms, wolf-raven cooperative hunting, multi-agent system, robust control, disturbance rejection.

I. INTRODUCTION

THE distributed cooperation of ground and aerial vehicles and robots constitutes an interesting class of systems with various applications in engineering, transportation, manufacturing, rescue contexts, mapping, etc. Hitherto, the cooperation of ground and aerial vehicles has been investigated following several directions. First, they have been studied for route searching and path planning, e.g. [1], [2]. Second, ground vehicles have been employed in missions of payload transportation and agile manipulation, which are guided by aerial vehicles to **avoid obstacles while keeping a rigid formation**, e.g. [3], [4]. Ground-aerial vehicle cooperation has also been investigated in construction missions [5], agricultural applications [6]. Third, aerial and ground vehicles have been researched to cooperate in other missions, for instance, fire detection and extinguishing [7], reconnaissance [8], surveillance [9], aerial vehicle landing on ground vehicle [10], etc. For a review of

ground-aerial vehicle cooperation problems, interested readers can take a look at [11] and [12].

The cooperation problem of ground and aerial vehicles to trace a moving ground object can be found in several realistic applications, e.g. surveillance, criminal tracing and military contexts, etc. In [13], the cooperative tracking of a moving ground object was studied but only for a group of aerial vehicles and without any ground vehicles. A similar problem was considered in [14] using nonlinear model predictive control. In [15], a study was made on the autonomous collaboration between a ground robot and a drone for mock-up disaster situations, where the ground robot was guided to navigate by the drone. In another work [16], ground and aerial vehicles collaborate for autonomous navigation to collect data in construction sites.

Due to the naturally existing obstacles in the surrounding environment, the position and velocity sensing of other vehicles by each vehicle is subject to disturbances. In [17], external disturbances and parametric uncertainties have been investigated for ground-aerial vehicle cooperation, but only for aerial vehicles, not between ground and aerial vehicles. To deal with the aforementioned disturbances, path-planning algorithms were proposed in [18] for ground and aerial vehicles to track a ground vehicle moving in urban environments. A second-order Markov chain and a dynamic occupancy grid were used to model the motion of the target vehicle and its vision obstructed by surrounding buildings. Nevertheless, an assumption on predictable movement of the traced vehicle is needed, which is not valid in many practical scenarios.

It should be noted that the considering cooperative vehicle tracing problem looks like the well-known leader-follower consensus control problem, but they are actually not the same. In the latter problem, the same type of agents are usually considered, e.g. schools of fishes, groups of birds, groups of ground robots, etc., hence the same space of agents is considered, and the control objective is the convergence of the following agents to the leading agents. However, in the former problem different types of agents are studied, distinct spaces of agents exist, and the control purpose is to trace closely the targeted agent, unnecessarily to converge to it.

Inspired by the wolf-raven cooperative hunting behavior in nature, this paper proposes a novel distributed control design for ground-aerial vehicle cooperation to trace a moving ground vehicle, without predictable movement assumptions nor path-planning nor mapping approaches. Here, roles of ground and aerial vehicles are similar to that of wolves and

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ravens, respectively, while the traced object is similar to the prey. Vehicle dynamics is represented by the nonlinear Dubins car model [19] which is then linearized for controller design.

Detailed contributions of this paper are described below.

- Disturbances on the collaboration between aerial and ground vehicles including time-varying velocity of the traced vehicle are studied. No assumptions on predictable movement of the traced vehicle, no path-planning or mapping methods, are used.
- A distributed robust controller is proposed for suppressing effects of disturbances on the position sensing between ground and aerial vehicles. Conditions are derived for the controller parameter in both H_∞ and H_2 robust control scenarios.
- The proposed distributed controller is proved to be input-to-state stable (ISS) in presence of bounded vehicle input constraints.

The remainder of this paper is as follows. The aerial-ground vehicle cooperation problem and its mathematical model are introduced in Section II. Then the proposed distributed controller design in presence of disturbances on position sensing and its robustness conditions are presented in Section III. Next, a distributed control design under a bounded input constraint is proposed in Section IV. A numerical example is subsequently given in Section V to illustrate the effectiveness of the proposed robust control design. Finally, Section VI concludes the paper.

The following notations and symbols will be used in the paper. $\mathbb{R}$ and $\mathbb{R}^n$ are used to denote the set of real scalars and vectors. Moreover, $\mathbf{1}_n$ and I_n denote the $n \times 1$ vector and the $n \times n$ identity matrix, respectively. The notation $\text{diag}\{\}$ denotes diagonal or block-diagonal matrices. Lastly, $\mathcal{K}_\infty$ denotes the class of scalar function $\gamma(x) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ which is continuous, strictly increasing, unbounded, and $\gamma(0) = 0$; and $\mathcal{KL}$ denotes the class of scalar function $\gamma(x, t) : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $\gamma(\cdot, t) \in \mathcal{K}_\infty$ for each t and $\gamma(x, t) \searrow 0$ as $t \rightarrow \infty$.

II. MATHEMATICAL MODEL FOR GROUND-AERIAL VEHICLE COOPERATION

A. Inspiration From the Wolf-Raven Cooperative Hunting

The wolf-raven cooperative hunting behavior recently introduced in [20] is a very interesting behavior in nature. Usually, predators compete with each other to hunt their preys, but in a rare scenario a wolf and a raven – two predators – collaborate with each other for hunting down a hare, as illustrated in Figure 1.

In their cooperation, the wolf and the raven complement each other, where the wolf lacks of the vision of preys due to the surrounding environment, e.g. trees and bushes, while the raven does not have enough strength to grasp big preys. As such, the raven, with a better vision from above, will guide the wolf to the prey location, and the wolf, after hunting down the prey, will share it with the raven. This cooperation provides many implications for use in different research areas, e.g. energy, transportation, space exploration, etc.

An illustration of the mathematical modeling attempt in [20] for the wolf-raven cooperative hunting problem is given

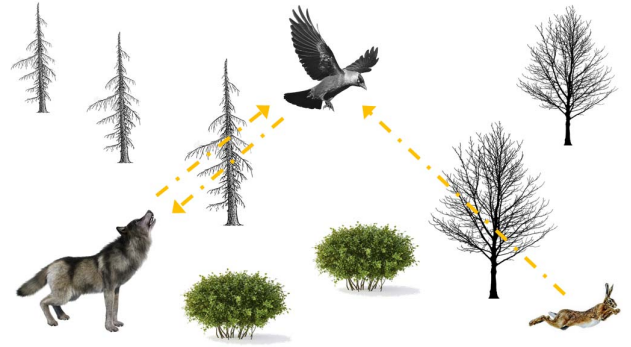


Fig. 1. Illustration for the wolf-raven cooperative hunting.

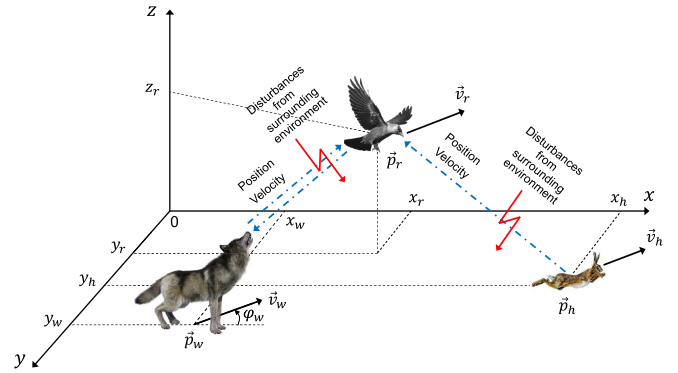


Fig. 2. Illustration for mathematical variables in the wolf-raven cooperative hunting.

in Figure 2, showing a similarity with the motion of front-wheel steering vehicles. This motivates the current work to exploit a vehicle cooperation problem in which vehicles moving in the air and on the ground collaborate for tracing a moving ground object, as depicted in Figure 3. Here, ground vehicles act similarly to the wolves and aerial vehicles hold the role of the ravens, whilst the traced ground vehicle is similar to the hare (prey).

Note that in their collaborative hunting, the raven and the wolf may not possess mapping or path-planning alike capabilities. Likewise, the tracing task of a ground vehicle may not be known a priori, and the traced vehicle movement can be uncertain and unpredictable, hence ground and aerial vehicles may not be able to perform any mapping or path-planning alike strategies. Therefore, no mapping or path-planning algorithms are considered in our current research. This distinguishes our study from other existing works in the literature as reviewed in the introduction section. It is also worth noting that the mathematical model of wolf-raven cooperative hunting in [20] was in form of a very general nonlinear dynamical system and no details were made there to study its features. Hence, the current work is not a simple extension of that in [20].

B. Vehicle Model

As shown in the literature (see e.g. [11], [21], [22], and others), the dynamic model of 4-wheel ground vehicles with front-wheel driving can be represented by the well-known

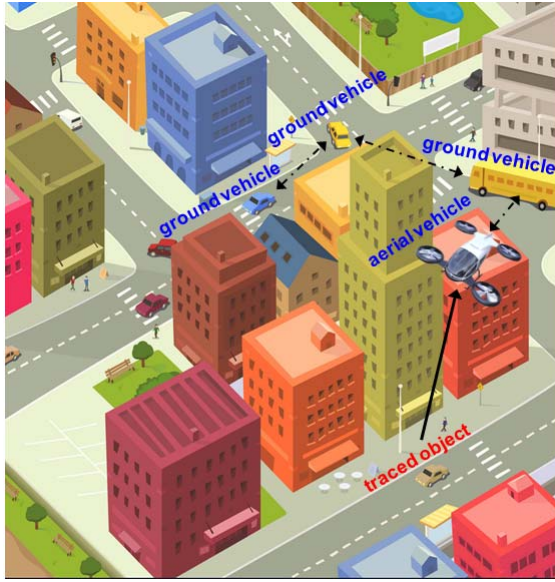


Fig. 3. Illustration of ground-aerial vehicle cooperation for tracing a moving object on the ground.

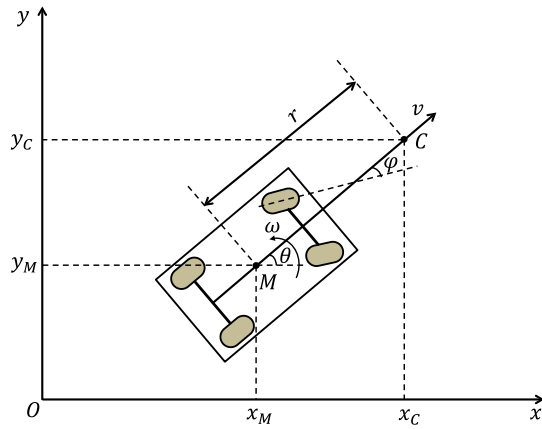


Fig. 4. Demonstration of variables defined for 4-wheel front-steering vehicles.

Dubins car model [19] to capture the essence of vehicle motions. The vehicle variables and parameters of each vehicle are shown in Figure 4, in which the vehicle center of mass is denoted by M whose position in a global coordinate frame (O, x, y) is represented by (x_M, y_M) . The rotating and steering angles are denoted by θ and φ , respectively; whereas ω and v represent the angular and longitudinal velocity. Then the mathematical model of a 4-wheel vehicle with front-wheel steering is represented by

$$\begin{aligned} \dot{\theta} &= \omega = v \tan \varphi, \\ \dot{x}_M &= v \cos \theta, \\ \dot{y}_M &= v \sin \theta. \end{aligned} \quad (1)$$

In this model, the control inputs for each vehicle are the steering angle φ and the longitudinal speed v , while the rotating angle θ is assumed to be measurable. Note that this Dubins car model can also be used for aerial vehicles in which their attitudes are assumed to be freely and independently controlled. This is in line with the considering context of aerial-ground vehicle cooperation, since we have assumed that

the aerial vehicle can freely vary its altitude depending on the surrounding environmental disturbances similarly to that of the raven in the wolf-raven cooperative hunting.

The Dubins vehicle model (1) can then be converted to a set of two independent integrators by a feedback linearization scheme [21], [22], in which the coordinate is changed from that on the center of mass M to that on a heading point C distanced by $r > 0$ from M on the longitudinal vector v . Denote $\bar{x} \triangleq \begin{bmatrix} x_C \\ y_C \end{bmatrix}$ and $\mathbb{M} \triangleq \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}$. We have

$$\dot{\bar{x}} = \mathbb{M}[v, \omega]^T. \quad (2)$$

Let

$$\bar{u} \triangleq \mathbb{M}[v, \omega]^T,$$

then the vehicle model can be rewritten in terms of the coordinates of C as a set of two integrators as follows,

$$\dot{\bar{x}} = \bar{u}. \quad (3)$$

Next, we have

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \mathbb{M}^{-1} \bar{u} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta / r & \cos \theta / r \end{bmatrix} \bar{u}. \quad (4)$$

Denote

$$\begin{aligned} \tilde{u}_1 &\triangleq \bar{u}_1 \cos \theta + \bar{u}_2 \sin \theta, \\ \tilde{u}_2 &\triangleq -\bar{u}_1 \sin \theta / r + \bar{u}_2 \cos \theta / r, \end{aligned} \quad (5)$$

where $\bar{u}_1$ and $\bar{u}_2$ are first and second elements of $\bar{u}$, respectively. It then follows from (4) that

$$\begin{aligned} v &= \tilde{u}_1, \\ \varphi &= \arctan \frac{\omega}{v} = \arctan \frac{\tilde{u}_2}{\tilde{u}_1}. \end{aligned} \quad (6)$$

Equation (3) shows that **vehicles can be independently controlled along plane axes**, and equation (6) shows how to compute actual control inputs, the longitudinal velocity v and the steering angle φ , for the original nonlinear model (1) of vehicles. Thus, in the current work we first develop a distributed control algorithm for a set of independent integrators representing the linearized model of vehicles (3) along each plane axis, and then reveal its robustness.

Now, suppose that there are one aerial vehicle and n ground vehicles collaborating to trace a moving ground vehicle. For convenience, the aerial vehicle is indexed 1, while ground vehicles are indexed from 2 to $n+1$. Then let $x_i(t)$ be the position of vehicle i in one plane axis, $i = 1, \dots, n+1$, and $x_o(t)$ and $v_o(t)$ be the position and speed of the traced ground vehicle, respectively; t is the time index. The information on $x_o(t)$ and $v_o(t)$ is only sensed by the aerial vehicle, while the aerial vehicle's position is sensed by some of ground vehicles, but not all of them.

Denote $x(t) \triangleq [x_1(t), \dots, x_{n+1}(t)]^T$. Then the linearized model for the considering group of aerial and ground vehicles along one plane axis is as follows,

$$\dot{x}(t) = u(t), \quad (7)$$

where $u(t)$ is the vector of control inputs.

Remark 1: Although the Dubins car model (1) is widely used in the literature, it does not take into account more detailed and complicated dynamics of vehicles including that of mechanical and electronic parts. More complete and complex models of vehicles will be investigated in future works.

C. Inter-Vehicle Communication and Sensing Graph

The inter-vehicle communication and sensing structure is described by an undirected graph $\mathcal{G}$ with $n + 1$ vertices. The vertex and edge sets of $\mathcal{G}$ are then denoted by $\mathcal{V}$ and $\mathcal{E}$, respectively. Each vertex represents a vehicle, while each edge $(i, j) \in \mathcal{E}$ indicates the communication and sensing link between vehicle j and vehicle i .

Denote $\mathcal{N}_i \triangleq \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$ the neighboring set of vehicle i . The adjacency matrix $\mathcal{A}$ of $\mathcal{G}$ consists of elements a_{ij} , where $a_{ij} > 0$ if $(i, j) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. The degree of each vertex is hence represented by $d_i \triangleq \sum_{j \in \mathcal{N}_i} a_{ij}$, by which we can define the degree matrix $\mathcal{D} = \text{diag}\{d_i\}_{i=1, \dots, n+1}$. Finally, the Laplacian matrix, denoted by $\mathcal{L}$, is $\mathcal{L} = \mathcal{D} - \mathcal{A}$.

III. DISTRIBUTED ROBUST CONTROLLER DESIGN

Let $w_i(t)$ and $w_o(t)$ represent the disturbances affected on the position sensing of vehicle i and the traced ground vehicle. Then the following distributed cooperative tracing controller is employed by aerial and ground vehicles along each plane axis,

$$u_i(t) = -\mu \sum_{j \in \mathcal{N}_i} a_{ij} [x_i(t) - x_j(t) + w_i(t) - w_j(t)] - \mu \sigma_i [x_i(t) - x_o(t) + w_i(t) - w_o(t)], \quad (8)$$

where $\mu > 0$ is a controller parameter, $\sigma_i = 1$ if vehicle i can sense the position of the traced vehicle, and $\sigma_i = 0$ otherwise. Since we have assumed that only the aerial vehicle can sense the traced vehicle's position, only $\sigma_1 = 1$ whereas $\sigma_i = 0$ for $i = 2, \dots, n + 1$. Denote $\Sigma \triangleq \text{diag}\{\sigma_i\}_{i=1, \dots, n+1}$, $d_1(t)$ and $d_2(t)$ the vectors of all $w_i(t)$ and $w_i(t) - w_o(t)$ in (8), respectively. Then Eq. (8) can be rewritten as follows,

$$u(t) = -\mu \mathcal{L}[x(t) + d_1(t)] - \mu \Sigma[x(t) - \mathbf{1}_{n+1}x_o(t) + d_2(t)]. \quad (9)$$

Next, let us denote the **tracing error vector** by

$$\zeta(t) \triangleq x(t) - \mathbf{1}_{n+1}x_o(t).$$

Then the controller (9) is rewritten as follows,

$$u(t) = -\mu(\mathcal{L} + \Sigma)\zeta(t) - \mu \mathcal{L}d_1(t) - \mu \Sigma d_2(t), \quad (10)$$

due to the fact that $\mathcal{L}\mathbf{1}_{n+1} = 0$. As a result, the closed-loop system model becomes,

$$\dot{x}(t) = -\mu(\mathcal{L} + \Sigma)\zeta(t) - \mu \mathcal{L}d_1(t) - \mu \Sigma d_2(t). \quad (11)$$

This leads to

$$\dot{\zeta}(t) = -\mu(\mathcal{L} + \Sigma)\zeta(t) - \mu \mathcal{L}d_1(t) - \mu \Sigma d_2(t) - \mathbf{1}_{n+1}v_o(t). \quad (12)$$

Let $y = \alpha \zeta$ be the output with $\alpha > 0$, and define

$$\mathcal{A} \triangleq -\mu(\mathcal{L} + \Sigma), \quad \mathcal{B} \triangleq [-\mu \mathcal{L}, \quad -\mu \Sigma, \quad -\mathbf{1}_{n+1}], \\ \mathcal{C} \triangleq \alpha \mathbf{I}_{n+1}, \quad d(t) \triangleq [d_1(t), \quad d_2(t), \quad v_o(t)]^T.$$

Then the robust control problem investigated in this paper is described in the following section.

Remark 2: The model of disturbances affected to vehicles considered in (8) is fairly simple. In realistic situations, the surrounding environment characteristics can be significantly varied, for example due to moving obstacles, different conditions from one location to another, etc. Hence, modeling of effects caused by disturbances from realistic surrounding environments to vehicle motions is challenging. Even simulating them in closer-to-reality systems is not commonly available or accessible. Thus, we stick to (8) and accordingly design a distributed robust controller to suppress the effects of modeled disturbances that will be validated via numerical simulations later.

A. Control Design Problem

Robust Control Problem: Given a disturbance suppression index $\gamma > 0$ and a bounded disturbance vector $d(t)$, design a distributed robust controller (9) for the cooperation of ground and aerial vehicles described in (7) such that

$$\|\mathcal{C}(s\mathbf{I}_{n+1} - \mathcal{A})^{-1}\mathcal{B}\| < \gamma. \quad (13)$$

The system norm in (13) can be system ∞ -norm or 2-norm. The condition (13) means that the effects of disturbances from the surrounding environment and the changing speed of the traced object to the tracing of vehicles can be suppressed to a level γ .

In the following subsections, we derive conditions under which the controller (9) is a distributed robust H_∞ controller and a distributed robust H_2 controller. Therefore, the controller (9) can help suppress the effects of different disturbance types to the cooperation of aerial and ground vehicles when tracing a moving ground object.

B. Distributed Robust H_∞ Control

In order to solve the robust control design problem introduced in Section III-A above as **the disturbances have finite energy**, H_∞ control design is needed, where the following basic lemma is utilized.

Lemma 1: Consider a linear system with transfer function $G(s) = \mathcal{C}(s\mathbf{I} - \mathcal{A})^{-1}\mathcal{B}$ with a Hurwitz matrix $\mathcal{A}$. Then given a $\gamma > 0$, the system H_∞ norm satisfies $\|G(s)\|_\infty < \gamma$ if and only if there exists a matrix $\mathcal{P} > 0$ which is the solution of the following Riccati inequality,

$$\mathcal{P}\mathcal{A} + \mathcal{A}^T\mathcal{P} + \gamma^{-2}\mathcal{P}\mathcal{B}\mathcal{B}^T\mathcal{P} + \mathcal{C}^T\mathcal{C} < 0. \quad (14)$$

Using Lemma 1 above for our considering problem with $\alpha = 1$, i.e., $\mathcal{C} = \mathbf{I}_{n+1}$, the robust cooperative vehicle tracing performance (13) is guaranteed if and only if there exists a $\mathcal{P} > 0$, $\mathcal{P} \in \mathbb{R}^{(n+1) \times (n+1)}$ such that

$$\mathcal{P}\mathcal{A} + \mathcal{A}^T\mathcal{P} + \gamma^{-2}\mathcal{P}\mathcal{B}\mathcal{B}^T\mathcal{P} + \mathbf{I}_{n+1} < 0. \quad (15)$$

Subsequently, the existence of a distributed robust H_∞ controller for solving the described design problem is shown in the following theorem.

Theorem 1: For a connected graph $\mathcal{G}$ and a $\mu > 0$, the controller (9) is a distributed robust H_∞ controller for the system (12) if the disturbance suppression γ satisfies the following condition,

$$\gamma > \max_{k=2,\dots,n+1} \left\{ 1 + \frac{1}{\mu^2}, \sqrt{\frac{1 + \lambda_k^2}{1 + 2\lambda_k}} \right\}. \quad (16)$$

Proof: Recall that for a connected undirected graph $\mathcal{G}$, $-(\mathcal{L} + \Sigma)$ is a Hurwitz matrix (see [23] and references therein). Then the proof is completed as long as we can find a $\mu > 0$ such that (15) is satisfied for some $\mathcal{P} > 0$.

Let $V \in \mathbb{R}^{(n+1) \times (n+1)}$ be the matrix of normalized eigenvectors of $\mathcal{L}$, then $\mathcal{L} = V\Lambda V^T$, where $\Lambda = \text{diag}\{\lambda_k\}_{k=1,\dots,n+1}$ with λ_k being the k -th smallest eigenvalue of $\mathcal{L}$. Consequently, we choose $\mathcal{P} = V\tilde{P}V^T$ where $\tilde{P} > 0$ and $\tilde{P}$ is diagonal. Then multiplying V^T and V to left and right of (15) gives us

$$-\mu\tilde{P}V^T(\mathcal{L} + \Sigma)V - \mu V^T(\mathcal{L} + \Sigma)V\tilde{P} + I_{n+1} + \gamma^{-2}\tilde{P}V^T(\mu^2\mathcal{L}^2 + \mu^2\Sigma + \mathbf{1}_{n+1}\mathbf{1}_{n+1}^T)V\tilde{P} < 0, \quad (17)$$

since $V^TV = I_{n+1}$ and $\Sigma^2 = \Sigma$. Moreover, we have $V^T\mathcal{L}V = \Lambda$ and $V^T\mathcal{L}^2V = \Lambda^2$, $V^T\mathbf{1}_{n+1} = e_1$, where e_1 is the basis vector with only the first element equal to 1 while all other elements are 0. Hence, (17) is equivalent to

$$-\mu\tilde{P}(\Lambda + V^T\Sigma V) - \mu(\Lambda + V^T\Sigma V)\tilde{P} + \gamma^{-2}\tilde{P}(\mu^2\Lambda^2 + \mu^2V^T\Sigma V + e_1e_1^T)\tilde{P} + I_{n+1} < 0. \quad (18)$$

Because $\Sigma = \text{diag}\{1, 0, \dots, 0\}$, we can easily compute $V^T\Sigma V$ to be $\mathbf{1}_{n+1}\mathbf{1}_{n+1}^T/(n+1)$. Therefore, (18) is rewritten as follows,

$$\begin{aligned} & -2\text{diag}\{\mu^{-1}\lambda_k p_k\} + \gamma^{-2}\text{diag}\{\lambda_k^2 p_k^2\} \\ & + 1/(n+1)[- \mu^{-1}(\mathbf{1}_{n+1}p^T + p\mathbf{1}_{n+1}^T) + \gamma^{-2}pp^T] \\ & + \mu^{-2}I_{n+1} + p_1^2\gamma^{-2}\mu^{-2}e_1e_1^T < 0, \end{aligned} \quad (19)$$

where p_k is the k -th element on the diagonal of $\tilde{P}$, and p is the vector of all p_k .

Let $\tilde{P}$ be κI_{n+1} , $\kappa > 0$, then (19) reduces to

$$\begin{aligned} & \sum_{k=2}^{n+1} [\mu^2\kappa^2\lambda_k^2 - 2\mu\kappa\lambda_k\gamma^2 + \gamma^2 + \kappa\mu(\kappa\mu - 2\gamma^2)]x_k^2 \\ & + [\kappa^2 + \gamma^2 + \kappa\mu(\kappa\mu - 2\gamma^2)]x_1^2 < 0, \end{aligned} \quad (20)$$

A sufficient condition for attaining (20) is that

$$\begin{aligned} & \mu^2\kappa^2\lambda_k^2 - 2\mu\kappa\lambda_k\gamma^2 + \gamma^2 + \kappa\mu(\kappa\mu - 2\gamma^2) < 0, \\ & \kappa^2 + \gamma^2 + \kappa\mu(\kappa\mu - 2\gamma^2) < 0, \end{aligned} \quad (21)$$

for all $k = 2, \dots, n+1$. By choosing $\mu\kappa$ to be 1, (21) becomes

$$\begin{aligned} & \gamma^2 > \frac{1 + \lambda_k^2}{1 + 2\lambda_k}, \\ & \gamma^2 > \kappa^2 + 1, \end{aligned} \quad (22)$$

for all $k = 2, \dots, n+1$. Then substituting κ by $\frac{1}{\mu}$, (22) is equivalent to (16). ■

If the disturbance suppression level γ is known or set a priori, then the controller parameter μ can be straightforwardly determined based on the result of Theorem 1 as follows.

Corollary 1: Given a connected graph $\mathcal{G}$ and a disturbance suppression level

$$\gamma > \max_{k=1,\dots,n+1} \left\{ 1, \sqrt{\frac{1 + \lambda_k^2}{1 + 2\lambda_k}} \right\}, \quad (23)$$

(9) is a distributed robust H_∞ controller for the system (12) if

$$\mu > \frac{1}{\sqrt{\gamma - 1}}. \quad (24)$$

A consequence of Corollary 1 is that the disturbance effect to the ground vehicle tracing performance will be smaller as the controller parameter μ is bigger.

C. Distributed Robust H_2 Control

To resolve the robust control design problem described in Section III-A when disturbances have infinite energy, this section provides an H_2 controller design. The following basic lemma is employed.

Lemma 2: Consider a linear system with transfer function $G(s) = C(sI - A)^{-1}B$ with a Hurwitz matrix A . Then given a $\gamma > 0$, the system H_2 norm satisfies $\|G(s)\|_2 < \gamma$ if and only if there exists a matrix $P > 0$ which is the solution of the following matrix inequalities,

$$AP + PA^T + BB^T < 0, \text{ trace}(C^T PC) < \gamma^2. \quad (25)$$

Utilizing Lemma 2 for our problem, the robust H_2 cooperative vehicle tracing performance (13) is guaranteed if and only if there exists a $\mathcal{P} > 0$, $\mathcal{P} \in \mathbb{R}^{(n+1) \times (n+1)}$ such that

$$\mathcal{A}\mathcal{P} + \mathcal{P}\mathcal{A}^T + \mathcal{B}\mathcal{B}^T < 0, \alpha^2 \text{trace}(\mathcal{P}) < \gamma^2. \quad (26)$$

Subsequently, the following theorem shows the existence of a distributed robust H_2 controller.

Theorem 2: For a connected graph $\mathcal{G}$ and a given $\alpha > 0$, if the disturbance suppression level γ satisfies

$$\gamma > \sqrt{n+1}\alpha, \quad (27)$$

then there exists a $\mu > 0$ such that the controller (9) is a distributed robust H_2 controller for the system (12).

Proof: The proof of this theorem is obtained in a similar manner as that for Theorem 1, where an abuse of the variables $\mathcal{P}$, V , $\tilde{P}$ and the notations p , e_1 is made here to avoid a burden on the number of variables and notations.

Multiplying V^T and V to left and right of (26) gives us

$$-\mu(\Lambda + V^T\Sigma V)\tilde{P} - \mu\tilde{P}(\Lambda + V^T\Sigma V) + \mu^2\Lambda^2 + \mu^2V^T\Sigma V + e_1e_1^T < 0. \quad (28)$$

This is equivalent to the following

$$\begin{aligned} & -2\mu\text{diag}\{\lambda_k p_k\} + \mu^2\text{diag}\{\lambda_k^2\} + e_1e_1^T \\ & + \mu/(n+1)[- (\mathbf{1}_{n+1}p^T + p\mathbf{1}_{n+1}^T) + \mu\mathbf{1}_{n+1}\mathbf{1}_{n+1}^T] < 0. \end{aligned} \quad (29)$$

If we choose $\tilde{\mathcal{P}}$ to be κI_{n+1} , $\kappa > 1$, then (29) becomes

$$\text{diag}\{1, \mu^2 \lambda_k^2 - 2\mu\kappa\lambda_k\}_{k=2,\dots,n+1} + \frac{\mu(\mu - 2\kappa)}{n+1} \mathbf{1}_{n+1} \mathbf{1}_{n+1}^T \prec 0. \quad (30)$$

By the Cauchy–Schwarz inequality, we have

$$x^T \mathbf{1}_{n+1} \mathbf{1}_{n+1}^T x \leq (n+1)x^T x,$$

for all $x \in \mathbb{R}^{n+1}$. Hence, the left hand side of (30), after being multiplied with x^T to the left and x to right, is not greater than

$$[\mu(\mu - 2\kappa) + 1]x_1^2 + \mu \sum_{k=2}^{n+1} [\mu\lambda_k^2 - 2\kappa\lambda_k + \mu - 2\kappa]x_k^2, \quad (31)$$

where x_k is the k -th element of x . The summand in (31) will be strictly negative for all $x \neq 0$ when $\mu\lambda_k < 2\kappa$ and $\mu(\mu - 2\kappa) + 1 < 0$. These two conditions lead to

$$\kappa - \sqrt{\kappa^2 - 1} < \mu < \min\{2\kappa/\lambda_{n+1}, \kappa + \sqrt{\kappa^2 - 1}\}.$$

Additionally, $\alpha^2 \text{trace}(\mathcal{P}) < \gamma^2$ with chosen $\mathcal{P}$ means $(n+1)\alpha^2\kappa < \gamma^2$. Under the condition (27), we can easily find κ in the interval $(1, \frac{\gamma^2}{(n+1)\alpha^2})$. Therefore, we can always find $\mathcal{P}$ and μ to satisfy (26). This completes the proof. ■

The following corollary shows a way to choose the controller parameter μ based on the proof of Theorem 2.

Corollary 2: Given a connected graph $\mathcal{G}$, a disturbance suppression level $\gamma > 0$, and an $\alpha > 0$ fulfilling condition (27), (9) is a distributed robust H_2 controller for the system (12) if μ satisfies

$$\kappa - \sqrt{\kappa^2 - 1} < \mu < \min\left\{\frac{2\kappa}{\lambda_{n+1}}, \kappa + \sqrt{\kappa^2 - 1}\right\}, \quad (32)$$

where κ is a parameter such that $1 < \kappa < \frac{\gamma^2}{(n+1)\alpha^2}$.

As seen in Corollary 2, the admissible range of the controller parameter μ is defined by λ_{n+1} and κ . Thus, given a communication and sensing graph $\mathcal{G}$, i.e. λ_{n+1} is fixed, the range for μ can be widened by increasing κ . One reasonable choice for α is a scaled value of $1/(n+1)$, for which the robustness of the averaged tracking error of aerial and ground vehicles to the moving ground object is assessed under inter-vehicle and sensing disturbances.

IV. TRACKING CONTROL UNDER BOUNDED INPUT CONSTRAINT

Due to physical limitations of realistic vehicles, their control inputs are subject to be bounded. Therefore, this section investigates the following constraint in the controller design.

• Input constraint:

$$u_i^{\min} \leq u_i(t) \leq u_i^{\max} \quad \forall i = 1, \dots, n+1, \quad \forall t > 0, \quad (33)$$

where $u_i^{\min}$ and $u_i^{\max}$ are known real constants for vehicle i .

Define the following saturation function,

$$\text{sat}(u_i) \triangleq \begin{cases} u_i^{\max} & \text{if } u_i > u_i^{\max}, \\ u_i & \text{if } u_i^{\min} \leq u_i \leq u_i^{\max}, \\ u_i^{\min} & \text{if } u_i < u_i^{\min}. \end{cases} \quad (34)$$

Then the distributed controller (10) under input bounded constraints becomes,

$$\begin{aligned} y(t) &= -\mu(\mathcal{L} + \Sigma)\zeta(t) - \mu\mathcal{L}d_1(t) - \mu\Sigma d_2(t), \\ u(t) &= \text{sat}(y(t)), \end{aligned} \quad (35)$$

where $\text{sat}(\cdot)$ is the vector of scalar saturation functions defined in (34). Next, we define

$$\begin{aligned} \phi(y_i) &\triangleq \text{sat}(y_i) - y_i, \quad i = 1, \dots, n+1, \\ \Phi(y) &\triangleq [\phi(y_1), \dots, \phi(y_{n+1})]^T = \text{sat}(y(t)) - y(t). \end{aligned} \quad (36)$$

The closed-loop tracing error network model becomes

$$\begin{aligned} \dot{\zeta}(t) &= y(t) + \Phi(y(t)) - \mathbf{1}_{n+1}v_0(t), \\ y(t) &= -\mu(\mathcal{L} + \Sigma)\zeta(t) - \mu\mathcal{L}d_1(t) - \mu\Sigma d_2(t). \end{aligned} \quad (37)$$

Let $Q \triangleq [-\mu\mathcal{L} - \mu\Sigma - \mathbf{1}_{n+1}]$, the system (37) can be rewritten as follows,

$$\dot{\zeta}(t) = -\mu(\mathcal{L} + \Sigma)\zeta(t) + Qd(t) + \Phi(y(t)). \quad (38)$$

Then some definitions related to the ISS are introduced below.

Definition 1: A dynamical system $\dot{x} = f(x, u)$ is said to be input-to-state stable (ISS) [24] if there exist some functions $\gamma_1 \in \mathcal{KL}$ and $\gamma_2 \in \mathcal{K}_\infty$ such that

$$\|x(t)\|_2 \leq \gamma_1(\|x(0)\|_2, t) + \gamma_2(\|u(t)\|_\infty). \quad (39)$$

Definition 2: Given a dynamical system $\dot{x} = f(x, u)$, $x(t) \in \mathbb{R}^n$, $x(0) = x_0$. A smooth function $V : \mathbb{R}^n \rightarrow \mathbb{R}_+$ is called an ISS-Lyapunov function [25] if: (i) there exist functions $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$ such that $\alpha_1(\|x\|) \leq V(x) \leq \alpha_2(\|x\|) \quad \forall x \in \mathbb{R}^n$; and (ii) there exist a function $\alpha : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that if $\|x\| \geq \chi(\|u\|)$ then $\dot{V}(x) \leq -\alpha(\|x\|)$.

The main result in this section is stated in the following theorem.

Theorem 3: For any given $\epsilon > 0$, the closed-loop system (38) is ISS with respect to the disturbance vector $d(t)$ and the additional input $\Phi(y(t))$, if μ satisfies

$$\mu > \frac{\epsilon}{2\lambda_{\min}(\mathcal{L} + \Sigma)}. \quad (40)$$

Proof: Consider a Lyapunov function $V(\zeta(t)) = \zeta(t)^T \zeta(t)$. For any $\epsilon > 0$, we have

$$\begin{aligned} \dot{V}(\zeta) + \epsilon V(\zeta) &= \zeta^T [-2\mu(\mathcal{L} + \Sigma) + \epsilon I_{n+1}] \zeta \\ &\quad + 2\zeta^T \Phi(y) + 2\zeta^T Qd. \end{aligned} \quad (41)$$

Denote $\tilde{\mathcal{L}} \triangleq -2\mu(\mathcal{L} + \Sigma) + \epsilon I_{n+1}$, then (41) is equivalent to,

$$\begin{aligned} \dot{V}(\zeta) + \epsilon V(\zeta) &= [\zeta - \tilde{\mathcal{L}}^{-1}\Phi(y) - \tilde{\mathcal{L}}^{-1}Qd]^T \tilde{\mathcal{L}} [\zeta - \tilde{\mathcal{L}}^{-1}\Phi(y) - \tilde{\mathcal{L}}^{-1}Qd] \\ &\quad - [\Phi(y) + Qd]^T \tilde{\mathcal{L}}^{-1} [\Phi(y) + Qd]. \end{aligned} \quad (42)$$

Since $-(\mathcal{L} + \Sigma)$ is a Hurwitz matrix, as seen in the proofs of previous theorems in Section III, we can always choose μ as in (40) such that $\tilde{\mathcal{L}}$ is also a Hurwitz matrix. Hence, we obtain

$$\begin{aligned} \dot{V}(\zeta) + \epsilon V(\zeta) &\leq -[\Phi(y) + Qd]^T \tilde{\mathcal{L}}^{-1} [\Phi(y) + Qd] \\ &\Leftrightarrow \dot{V}(\zeta) \leq -\epsilon V(\zeta) - [\Phi(y) + Qd]^T \tilde{\mathcal{L}}^{-1} [\Phi(y) + Qd] \end{aligned} \quad (43)$$

As such, $V(\zeta)$ is an ISS-Lyapunov function, and therefore, the closed-loop system (37) is ISS [25]. ■

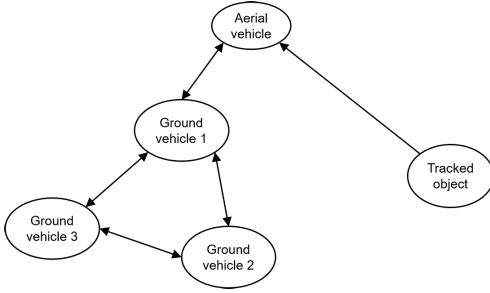


Fig. 5. Sensing graph in the numerical example.

V. NUMERICAL EXAMPLE

This section aims at illustrating our proposed distributed robust controller designs for the cooperation of aerial and ground vehicles to trace a moving ground object. For illustration purpose, one aerial vehicle and three ground vehicle are considered, i.e. $n = 3$, but it is readily scalable for higher numbers of vehicles. The inter-vehicle communication and sensing structure is illustrated in Figure 5. For simplicity, the non-zero elements of the adjacency matrix A associated with this communication graph, i.e. the edge weights, are assumed to be 1.

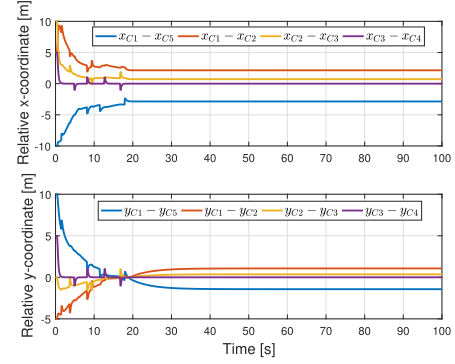
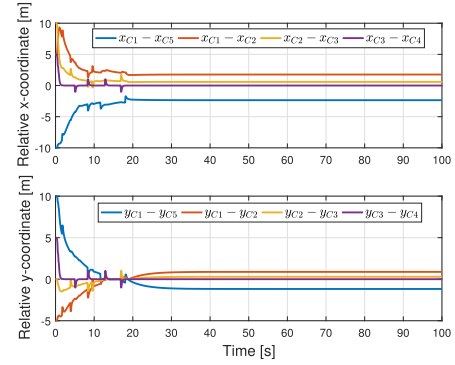
Initial conditions of the vehicles are as follows. For the traced ground vehicle, $x_{C0} = 35$, $y_{C0} = -5$, and $\theta_0 = 2\pi/3$, whereas the aerial vehicle has $x_{C1} = 25$, $y_{C1} = 5$, and $\theta_1 = 1.5\pi/3$. The attitude of the aerial vehicle is not considered in the simulation, because we previously assumed in Section II-B that it can be freely changed. For the other three ground vehicles, $x_{C2} = 20$, $y_{C2} = 10$, and $\theta_2 = 0$; $x_{C3} = 10$, $y_{C3} = 10$, and $\theta_3 = 0$; $x_{C4} = 5$, $y_{C4} = 5$, and $\theta_4 = \pi/3$.

In the simulations, the designed robust distributed controllers are first employed for the linearized models of vehicles. Consequently, control inputs for the original nonlinear models (1) of vehicles are computed as in (6). All simulations are carried out in Matlab and Simulink installed on a desktop computer having an Intel Core i7-6700K CPU 4GHz and a 64GB RAM.

It is assumed that movement of the traced ground vehicle is steered by step reference inputs. More specifically, its velocity step reference input is triggered after 5 seconds, while its steering angle step reference input is occurred at 15 seconds. These basically mean that the traced ground vehicle first moves straight and then switches its direction.

A. Illustration for Distributed Robust H_∞ Controller Design

Suppose that perturbations to the position sensing of other vehicles by each vehicle, which are caused by surrounding environments including building blocks, roadside objects, etc., can be represented by rectangular pulses. The triggering times and periods of those rectangular pulses are different between vehicles, although the amplitudes of all pulses are assumed to be 1 for simplicity. The on and off times are: 1 and 7 seconds for aerial vehicle; 3 and 8 seconds for ground vehicle 1; 7 and 15 seconds for ground vehicle 2; 4 and 11 for ground vehicle 3; 10 and 16 seconds for the traced ground vehicle.

Fig. 6. Cooperative tracing performance under distributed H_∞ control when μ is small ($\mu = 1.4$).Fig. 7. Cooperative tracing performance under distributed H_∞ control when μ is bigger ($\mu = 1.7$).

With the inter-vehicle communication and sensing graph defined in Figure 5, the disturbance suppression level γ should be greater than 1.3744, according to (23). Therefore, γ is set to be 1.3844 in the simulations. Subsequently, μ is selected to be $\mu = 1.4 < 1/\sqrt{\gamma - 1} = 1.613$, i.e. μ does not satisfy (24) in Corollary 1. Simulation results with this value of μ are shown in Figure 6. As seen, the relative positions of vehicles are mostly non-zero which imply that the tracing performance is not good.

Next, the value of μ is increase a bit to be 1.7, which induces simulation results in Figure 7. Tracing performance is then improved a little, where relative distance between vehicles are smaller than that in Figure 6.

To obtain a good tracing performance, μ is increased to be very big, in particular $\mu = 16$. Simulation results in this case are depicted in Figures 8–9. It can be seen from Figure 8 that aerial and ground vehicles now can asymptotically trace the dedicated ground vehicle, while suppressing the effect of disturbances. This can also be seen from the trajectories of all vehicles shown in Figure 9, hence illustrating the effectiveness of our proposed distributed robust H_∞ controller design in Theorem 1 and Corollary 1.

B. Illustration for Distributed Robust H_2 Controller Design

In this scenario, the position sensing between aerial and other ground vehicles are assumed to be affected by normally distributed disturbances with zero means and distinct variances: 0.3 for aerial vehicle; 0.2 for the 1st ground

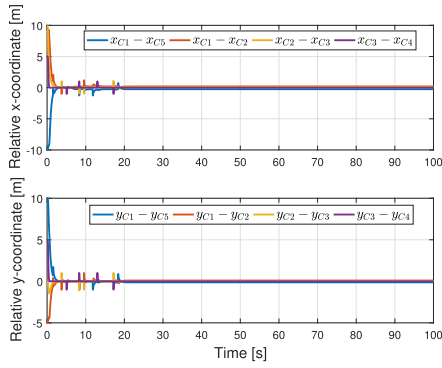


Fig. 8. Cooperative tracing performance under distributed H_∞ control when μ is very big ($\mu = 16$).

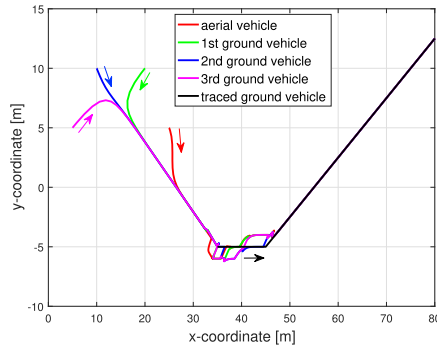


Fig. 9. Trajectories of vehicles under distributed H_∞ control when μ is very big ($\mu = 16$).

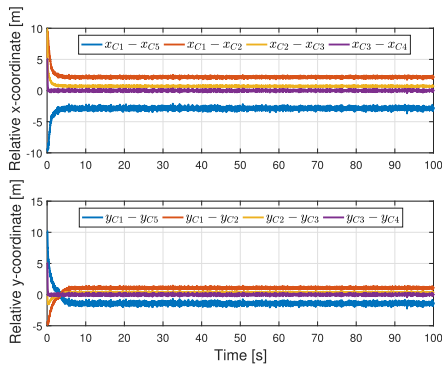


Fig. 10. Cooperative tracing performance under distributed H_2 control when μ is small ($\mu = 1.4$).

vehicle; 0.25 for the 2nd ground vehicle; and 0.15 for the 3rd ground vehicle. Lastly, the sensed information of the traced ground vehicle by the aerial vehicle is disturbed by a normally distributed disturbance with mean and variance equal to 0 and 0.1, respectively.

For a given disturbance suppression level γ , we first choose α so that $\frac{\gamma^2}{4\alpha^2} = 3$, and then select $\kappa = 2\sqrt{2}$, following Corollary 2. The condition (32) gives $\mu \in (0.1827, 1.4142)$, hence μ is chosen to be 1.4. Simulation results for original nonlinear models of vehicles are then depicted in Figures 10, showing that relative distances between vehicles in both x and y coordinates do not converge to 0. Moreover, the disturbance suppression is bad.

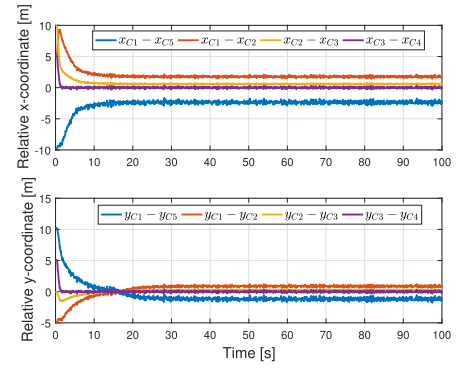


Fig. 11. Cooperative tracing performance under distributed H_2 control when μ is bigger ($\mu = 1.7$).

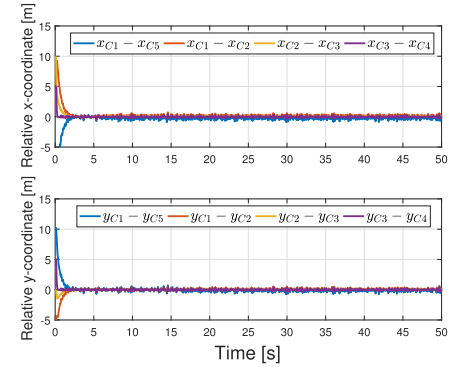


Fig. 12. Cooperative tracing performance under distributed H_2 control when μ is very big ($\mu = 16$).

Consequently, we verify the cooperative tracing of aerial and ground vehicles when $\frac{\gamma^2}{\alpha^2}$ is set to be bigger so that κ can be selected to be 3.6. Hence, $\mu \in (0.1417, 1.8)$, due to (32). Accordingly, μ is chosen to be $\mu = 1.7$. In this case, simulation results are shown in Figure 11, revealing that settling times are longer than that in the previous case depicted in Figure 10. However, the disturbance suppression is much better, as can be observed in Figure 7.

Next, the cooperative tracing performance is assessed when μ can be increased to be very big, says $\mu = 16$ (this also implies that κ should be very big). Then the tracing performance is exhibited in Figure 12, while trajectories of all vehicles are displayed in Figure 13. It is interesting to see that both settling times, tracing performance, and disturbance suppression are much better compared to previous cases shown in Figures 10–11. In particular, aerial and ground vehicles asymptotically converge to the traced ground vehicle, whereas the effect of disturbances is made to be very small. It is also interesting to see the differences on the disturbance suppression performance in Figure 13 and Figure 9, under the same controller parameter but different disturbance types.

C. Illustration for Tracing Control Under Bounded Input Constraints

In this section, we illustrate the performance of our proposed controller under saturated control input constraints. First of all, to see why such constraints are needed, we show how control inputs look like when no input constraints are enforced.

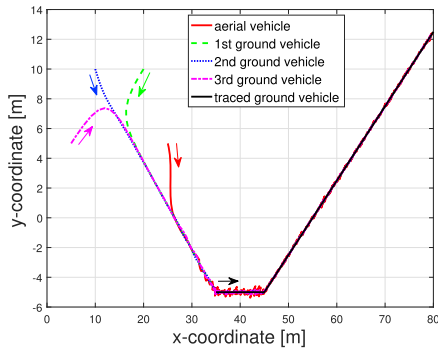


Fig. 13. Trajectories of vehicles under distributed H_2 control when μ is very big ($\mu = 16$).

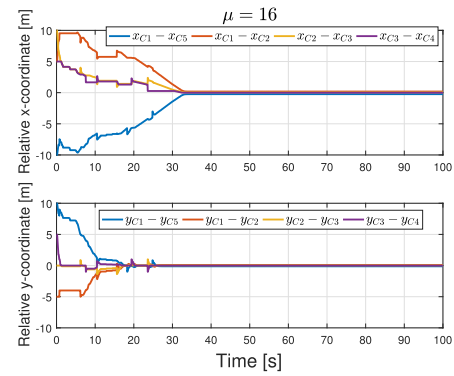


Fig. 16. Cooperative tracing performance under saturated control inputs.

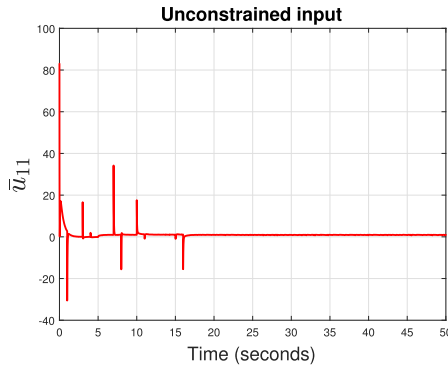


Fig. 14. A control input of aerial vehicle linearized model, as $\mu = 16$, without any input constraint.

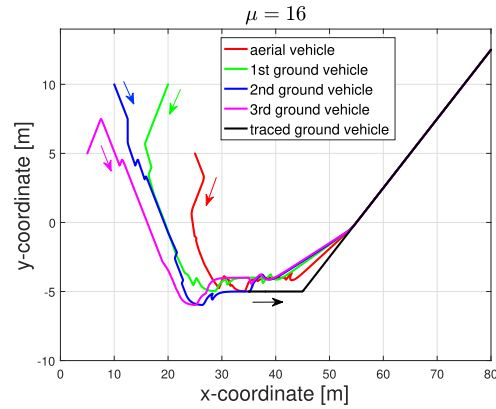


Fig. 17. Trajectories of vehicles under saturated control inputs.

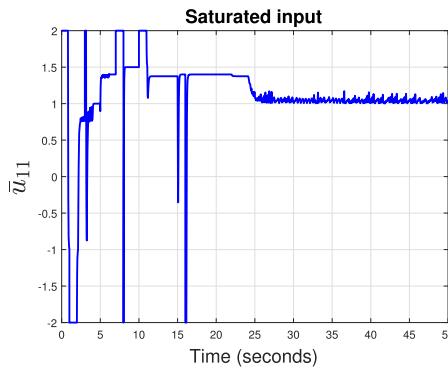


Fig. 15. Control input of aerial vehicle linearized model, as $\mu = 16$, under input saturation constraint.

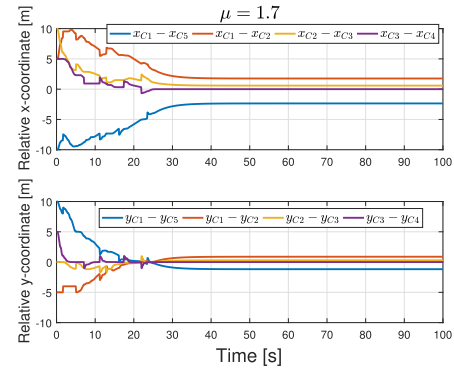


Fig. 18. Cooperative tracing performance under saturated control inputs and a smaller value of μ .

Particularly, Figure 14 depicts the first control input of the linearized model for the aerial vehicle, i.e. $\bar{u}_{11}$, at the same value of μ as in the context of Figures 8–9. It can be observed that this control input can be very large at several time instants, which are not possible in realistic systems.

Consequently, we assume that all individual control inputs of all vehicle linearized models are constrained to be within the interval $[-2, 2]$. Simulation results in this constrained input case are exhibited in Figures 15–17. Compared to that displayed in Figures 8–9, simulation results in Figures 16–17 show that the tracing time under input constraints is much longer than that without any constraints, although the tracking

is eventually achieved. This is logical, due to the limited control actions in presence of input saturation constraints.

Lastly, we show the tracing performance under input saturation constraints when the value of μ is smaller. For comparison purpose, $\mu = 1.7$ is selected, similarly to the scenario in Figure 11. Simulation results are then shown in Figure 18. Obviously, the tracing performance is much worse than that in case of bigger μ shown in Figure 16.

VI. CONCLUSION

This paper presents a novel control design approach for the problem of cooperatively tracing a moving ground object,

which is inspired by the wolf-raven cooperative hunting in nature. Accordingly, an aerial vehicle, which has a better vision of the traced object, is used to guide a group of ground vehicles, mimicking the roles of the raven and the wolfs, respectively, while the traced object is regarded similarly to the prey. The well-known Dubins car model is first utilized to describe the nonlinear dynamics of vehicles, which is then linearized to obtain a set of two integrators for each vehicle showing the independent control along each plane axis. Consequently, to cope with disturbances caused by the surrounding environment as well as by the time-varying velocity of the traced object, a distributed controller is proposed for the linearized model of vehicles. Next, conditions are derived such that the proposed distributed controller is robust in the H_∞ or H_2 sense depending on disturbance types. Based on those conditions, analytical and explicit bounds for the controller parameter are obtained. In the last theoretical part of this work, the bounded control input constraint is studied. It is proved that the input-to-state stability is guaranteed by the proposed controller, even in presence of disturbances and input saturation. Numerical simulations are then conducted to illustrate the performance of the proposed distributed robust controller design for all scenarios of H_∞ , H_2 , and input saturation control.

In the future works, more complicated models of vehicles taking into account the dynamics of different vehicle parts should be investigated to make the research closer to the reality. Another direction is to implement and further improve the derived results to more realistic systems having four-wheel ground vehicles (robots) and drones. Last but not least, more practical and sophisticated models of surrounding environment disturbances should be developed and taken into account in the control designs for the ground-aerial vehicle cooperation.

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